

# GLOBAL ISSUES IN ENGINEERING ETHICS

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DATA PRIVACY CONCERNS IN HUAWEI'S OPERATIONS



# INTRODUCTION

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- Globalization brings ethical challenges for tech companies.
- Huawei's data privacy practices raise global concerns.
- Focus: Engineering ethics and data privacy in Huawei.

# WHAT IS ENGINEERING ETHICS?

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- Protect public welfare, safety, and privacy.
- Ensure honesty, accountability, and transparency in technology.

# DATA PRIVACY:A GLOBAL ISSUE

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- Personal data is valuable and vulnerable.
- Engineers must protect data integrity and security.
- Data misuse leads to trust erosion and legal risks.

# HUAWEI OVERVIEW

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- Leading global telecom and tech company.
- Operations in over 170 countries.
- Criticized for alleged data privacy violations.

# ETHICAL CONCERNS WITH HUAWEI

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- Accusations of unauthorized data collection.
- Potential ties to state surveillance.
- Global debate over security and privacy.

# ENGINEERING RESPONSIBILITIES

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- Design secure systems protecting user data.
- Implement ethical AI and IoT data practices.
- Avoid hidden data flows and backdoors.



# GLOBAL DATA PRIVACY STANDARDS

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- GDPR (Europe): User consent, data minimization.
- CCPA (California): Consumer data rights.
- Huawei must align with international regulations.



# HUAWEI AND GDPR COMPLIANCE

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- Huawei claims compliance with GDPR.
- Concerns remain over data storage and transfer.
- Need for transparent data practices.

# SECURITY RISKS IN 5G NETWORKS

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- Huawei is a global 5G leader.
- Critics fear potential for espionage via infrastructure.
- Calls for independent audits and verification.

# ENGINEERING ETHICS: DATA PROTECTION

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- Confidentiality: Keep user data safe.
- Integrity: Prevent data tampering.
- Availability: Ensure data is accessible to rightful owners only.

# HUAWEI'S DATA HANDLING PRACTICES

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- End-to-end encryption in some services.
- Data localization efforts in compliance regions.
- Questions about backend data access remain.

# TRANSPARENCY AND TRUST

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- Transparency is key to public trust.
- Huawei's lack of open source scrutiny fuels skepticism.
- Need for verifiable ethical commitments.

# WHISTLEBLOWING AND ETHICAL DILEMMAS

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- Engineers face dilemmas if asked to compromise privacy.
- Whistleblowing can prevent harm but risks personal loss.
- Ethical culture is crucial.



# ROLE OF GOVERNMENT AND REGULATION

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- Governments must enforce privacy regulations.
- International cooperation needed for global companies.
- Ethical auditing as a regulatory requirement.



# CORPORATE SOCIAL RESPONSIBILITY (CSR)

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- CSR includes protecting digital rights.
- Huawei invests in social programs but faces ethical scrutiny.
- Balancing profit with ethical tech development.

# ETHICAL DESIGN AND DEVELOPMENT

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- Integrate **privacy by design** in all engineering phases.
- Continuous ethical training for engineers.
- Adopt global ethical guidelines for data handling.

# CASE STUDIES AND CONTROVERSIES

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- Huawei's ban in the US and some EU countries.
- Allegations of backdoor access in telecom equipment.
- Response: Huawei denies involvement in unethical practices.

# LESSONS LEARNED

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- Data privacy must be prioritized in engineering decisions.
- Global companies must adhere to ethical standards everywhere.
- Transparency and independent audits build trust.

# FUTURE OF DATA PRIVACY IN TECH

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- Growing need for data ethics in AI, IoT, and telecom.
- Privacy-centric innovation is essential.
- Global consensus on data ethics is emerging.

# CONCLUSION

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- Huawei's case highlights global ethical challenges in tech.
- Engineers must champion data privacy and ethical practices.
- Building trust requires ongoing commitment to ethics.